# A Review on Automated Plant Classification system using machine vision

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**Abstract**

Plants are one of the main sources of food, medicine and oxygen. There are nearly 350,000 plant species in the world. Plants comes in various sizes, with different leaf shapes, texture and edges and identifying all these plants is not an easy task for humans and might induce errors. Plant Classification through computer vision can be a method to reduce human errors and produce accurate results. There have been various studies on this problem and many research papers are published on the topic. In this paper we study 27 research papers and analyze the datasets and methods used in these papers. After summarize some findings observed through this analysis.

## 1. Introduction

Plants are an integral part of ecosystem. Plants are useful for human being and other living things. They are useful as foodstuff, as medicine and also in many industries. Identifying plants helps ensure the protection and survival of all-natural life. Plant identification is an important component of typical workflows in plant ecological research and making it accessible for non-experts would be highly beneficial. Even for researchers with expert knowledge on given plant, the process of manual plant species identification may be difficult to scale to high-throughput demands, while for non-experts, it may be prohibitively time consuming. Historically, identifying an unknown plant species required the consultation of a heavy field guide, where the user had to make sometimes obscure observations of the plant's features. The advent of highly-portable, computationally-powerful smart phones with large storage capacity presents an opportunity to not only replace and improve the database and decision tree functions of the field guide, but also to create automatic leaf classification applications based on well-known image processing methods currently becoming standardized on mobile platforms. A user-friendly application on a popular mobile platform such as Android and IOS that is capable of identifying a good number of plant species could achieve widespread adoption, increasing the public's knowledge of and appreciation for their environment.

This study summarizes about 27 different research papers dating from 2015 up till now on Plant Identification and its Classification. On scrutinizing all the papers, we came into knowledge about 5 common steps taken by the fellow authors. These are the generic steps that are mainly performed in any classification process. The steps include Dataset, Image Pre-processing, Feature Extraction, Classification, Deployment of the model.

- **Dataset:** When working with any Machine Learning, Computer Vision or Deep Learning project, one of the most crucial part is the dataset collection. Dataset is known to be the nutrition of any self-learning algorithm which makes it a crucial factor to maximize its performance.

- **Image Pre-processing and Data Augmentation:** Once the data is gathered, it is not always perfect for the application which it is intended for. Every data has some unwanted noise, different units and different

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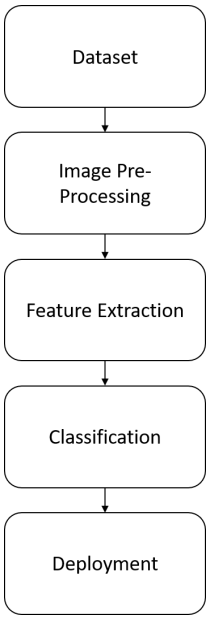


Figure 1: Five Generic Steps in any Classification Problem

sizes in case of image. Hence, Image pre-processing comes into the play. With the help of image pre-processing the dataset is made suitable for further steps.

- **Feature Extraction:** Feature extraction is a part of the dimensionality reduction process, in which, an initial set of the raw data is divided and reduced to more manageable groups. So, when you want to process it will be easier. The most important characteristic of these large data sets is that they have a large number of variables. These variables require a lot of computing resources to process them. So, Feature extraction helps to get the best feature from those big data sets by select and combine variables into features, thus, effectively reducing the amount of data.

- **Classification:** In this step, all the features extracted through feature extraction are grouped together and similar features are classified as one group. All this data is then fed into different classifiers which are based on Neural Networks, Machine Learning, etc. depending on the application.

- **Deployment of the model:** Dataset is gathered, it is cleaned, features are extracted and a classifier is also trained. But all these is done on an Integrated Development Environment. Which is not user friendly. These applications need to be deployed and a User Interface needs to be developed to make it presentable and useable by a layman. There are many ways to deploy a machine learning project including Android/iOS Application, Webservices, Desktop Application, etc.

## 2. Methods

This section gives a comparative analysis and a brief overview of the above mentioned five generic steps in the papers studied.

### *2.1. Dataset*

A dataset is collection of data. Data can be in any form i.e. Text, Numeric, audio, video, image, etc. Data is one of the important factors in the growth of Data Analysis/Machine Learning nowadays. The focus of this paper is image-based plant classification. So, we are concerned about the data in form of images. While working with any classification problem one of the most important component of the project is the dataset. Dataset is known to be the nutrition of any classification problem.

During any classification project dataset is divided into three parts, namely, training set, validation set and testing set. Training set, as the name suggests is used for training of any classifier/model. Validation set is used to keep an eye on the parameters of the model such as accuracy, error, etc. Some of the parameters of the models are calculated with validation set after several iterations. Once the training is completed, testing set comes into the picture which is used to evaluate the model's performance and its metrics.

There are various datasets available for plant classification. All the datasets are unique to themselves based on the method of photos taken. We classify them into two types based (Figure 2) on the organ of the plant present in the image, namely. The two types are Only Leaf and Multiple Organs. In the Only Leaf type datasets, the



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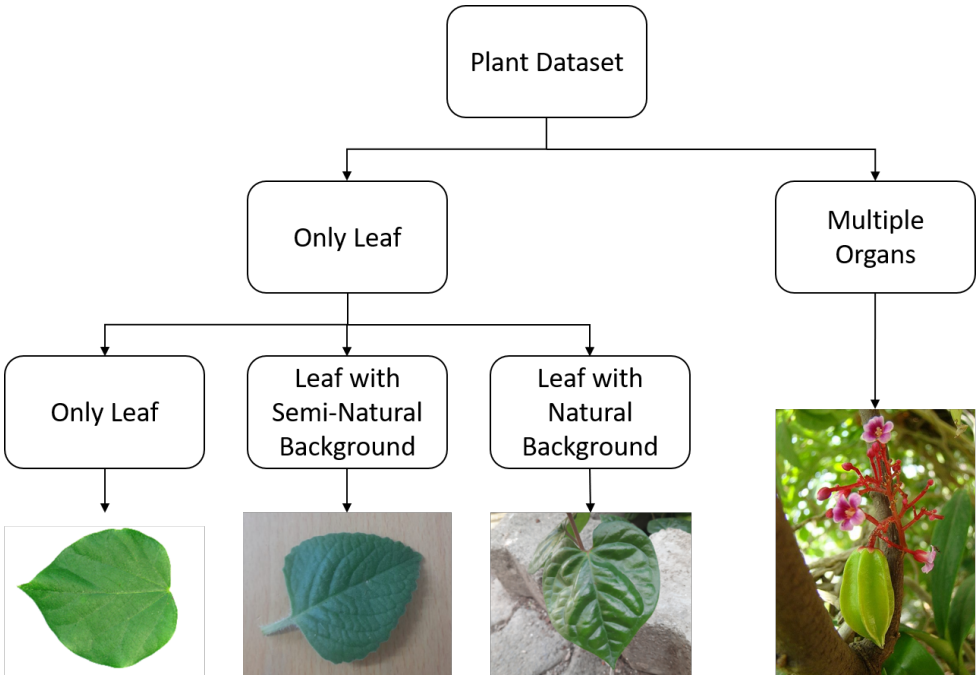


Figure 2: Tree Diagram of the Classification of Plant Dataset

main organ by which classification can be done is leaf. The dataset contains images with only leaves, images with leaves in natural background, images with leaves in semi-natural background. In the Multiple Organs type dataset, the classification is done through flowers, stems, leaves, etc.

There are multiple publicly available datasets that contains images of various types including only leaf, leaf with semi-natural background and with natural background. Some of the popular datasets are listed below:

- **Flavia Dataset** This dataset was developed by S. G. Wu [1] in their paper named "A Leaf Recognition Algorithm for Plant classification Using Probabilistic Neural Network". During the research they found out that there was almost no publicly available dataset to compare the algorithms developed in plant classification domain. Hence the dataset was developed. Flavia dataset contains 1907 images spread across 32 species. All the images are in the form of JPG. The images in this dataset falls under the category of Only Leaf.

- **ICL Leaf Dataset:** The Intelligent Computing Laboratory developed the ICL Leaf dataset which contains about 16,848 images spread across 220 plants. The images in this dataset falls under the category of Only Leaf.

- **Swedish Leaf Dataset:** The Swedish Leaf dataset contains leaf images from about 15 different species spread across Sweden. There are a total of 1125 images with approximately 75 samples of each species. The category for this dataset is Only Leaf.

- **Plants Seedling Dataset:** This dataset contains images of plant leaf at several growth stages. There is a total of 12 species present in the dataset with about 960 unique plants. The dataset was developed by Giselsson et. al. at Aarhus University Flakkebjerg Research station in a collaboration between University of Southern Denmark and Aarhus University. The imaes in this dataset falls under the category of Leaf with Natural Background.

After reviewing about 27 plant classification related papers, we classified them according to the above given classification in Figure 2. The most popular and widely used type was Only Leaf. Due to the rise in publicly available datasets many researchers used the same for comparison of their



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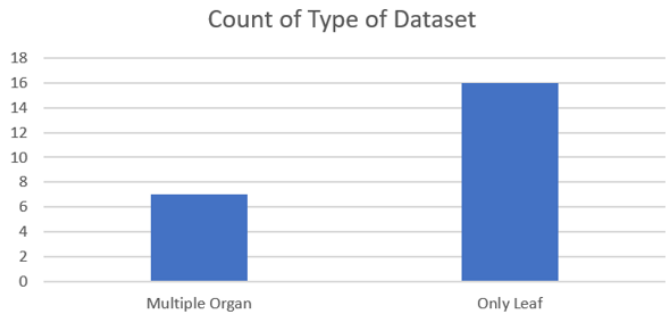


Figure 3: Count of Types of Dataset

proposed models, out of which the most widely used dataset was Flavia [2, 3, 4, 5, 6, 7, 8, 9, 10], Plant Seedling [11, 12, 13], Royal Botanic Garden [14, 4], ICL Leaf Dataset [15, 9]. Only 4 researches [16, 17, 18, 19] used the dataset with Multiple Organs.

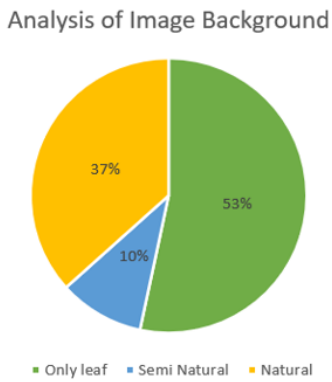


Figure 4: Pie Chart of Types of Image Background

The performance and classification of the model is proportional to the quality of the dataset used. A dataset should include all varieties of possibilities that can occur while using the application. Similarly, while deciding a plant classification dataset we must keep in mind that the model should be able to classify any type of image whether it be an image with natural, semi-natural background or an image with only leaf in it. The 27 papers that were analyzed by us had a variety of datasets used in them. The pie chart in figure 4 explains the proportion of types of images in the datasets we came across in this study.

### *2.2. Image Preprocessing*

Datasets are not always clean. There might be some different units which needs to be normalized. Data cleaning is the process of making a dataset suitable for an intended purpose by normalizing, resizing, removing unwanted parameters, etc. Uniformity is achieved by data cleaning. Similarly, for classification problem based on images, preprocessing is required. Image pre-processing involves resizing, color correction, color space conversion, orientation correction, etc. Data Augmentation is one of the most popular and rapidly growing technique used to avoid overfitting and enlarging the dataset. Data Augmentation in the field of image is a similar technique to Image preprocessing where data is generated by altering the orientation, size, color, horizontal and vertical shifting, image zooming, image flipping, rotation and other parameters to increase the size of the dataset.

Besides the wide advantages of Data Augmentation, it is still not widely used in the plant classification research. Belal A. M. Ashqar et. al. [11] used size re-scaling, rotations of 40 degrees, horizontal shift, image zooming, and horizontal flipping to increase the Plant Seedlings dataset images from 5608 to 9527. S. H. Lee [14] used rotation of images at 7 different angles i.e. 45 degrees, 90 degrees, 135 degrees, 180 degrees, 225 degrees, 270 degrees and 315 degrees. Josef Haupt et. al. [20] used Horizontal and vertical flip, zooming, rotating, shearing and shifting to increase the size of the dataset. Catherine R. Alimboyong et. al. [12] increased its dataset from 4234 images to 118750 images using data augmentation. Image Flipping, Image Rotation, Image Scaling, Image Flip-scale, Histogram equalization, RGB concentration modification was done to increase the dataset strength. Mads Dyrmann [21] increased the dataset by 5 times from 10,413 images to 50,864 images using image rotation and image mirroring.

Most of the papers classify different plants through an organ which can be leaf, flower, fruits, etc. To achieve this, image segmentation needs to be done. There are multiple techniques involved in image segmentation. Esraa Elhariri et. al. [22] uses a dataset of images of leaves having semi-natural background. They use morphological erosion of 5x5 structure to create a binary



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mask which is then multiplied with the original image to remove the background. David Knight et. al. [23] performs image segmentation on an image with Semi-natural background. G. Saleema et. al. [8] uses color space conversion for image segmentation. Since the dataset used is Flavia dataset with simple white background, it makes it easier for background removal process. The image is converted from RGB to L * a * b color space where L is the luminosity and 'a' and 'b' are color dependent variable. Aimen Aakif et. al. [9] uses a dataset with semi natural background which is converted into grayscale and then the image is binarized to remove the unwanted background from the image.

Apart from Data Augmentation and Image Segmentation, most of the papers utilize the conventional methods to make the dataset ready for feature extraction. Most common technique used are image resizing, image rescaling, grayscale conversion and contour detection.

### *2.3. Feature Extraction*

Image Preprocessing cleans all the images and creates a uniformity in the dataset. But they are still in the form of an image which is not an ideal form to feed the classifier model. They all need to be converted into a more suitable form where Feature Extraction comes into play. Feature Extraction is the process where data is converted into more easily and efficiently processable form. It helps to reduce the computing power required by reducing the large data.

Figure 5 shows the tree diagram of classification of Features i.e. Self-Extracted Features and Deep Learning Based Features. Anubha Pearline et. al. [7] compares both the methods for feature extraction on four publicly available datasets i.e. Folio, Flavia, Swedish Leaf Dataset, Leaf12 Dataset. In that study, 13 Hand Crafted features were calculated for all datasets and 4 Pre-Trained Models were used as feature extractors. In all four datasets the deep learning-based feature extractors yielded better results compared to the Self-Extracted Features.

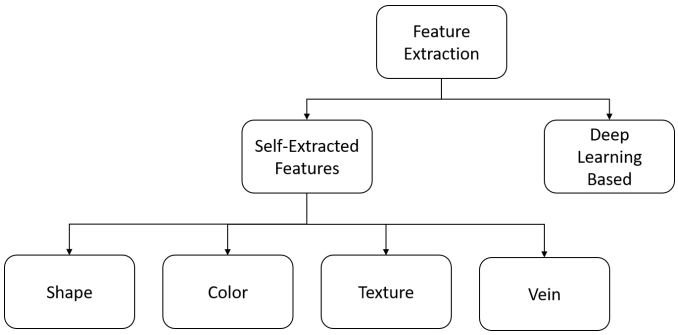


Figure 5: Types of Feature Extraction Techniques

#### *2.3.1. Self-Extracted Features*

Self-Extraction of features in the domain of image classification is the traditional and the most used way of extracting features. The fundamental features which can be extracted through this method are classified into 4 types, i.e. Shape, Color, Texture and Structural Features.

Shape was the most used feature category in the papers we reviewed. The shape of leaves varies with respect to their position of in the plant. Some of the common leaf shapes are Linear, Oblong, Elliptical, Ovate, Obovate, etc. These varieties of shape can be a very good differentiator of plants and will hold a high weight in any classifier. Leaves has different types of margin which are Smooth margin, Toothed margin, Lobed margin and Parted margin.

These parameters can be used to generate various features for machine learning algorithms such as rectangularity, eccentricity, aspect ratio, leaf arrangement, maximal indentation depth and many more. Esraa Elhariri et. al. [22] used 8 different shape features which includes Eccentricity, Solidity, Aspect Ratio, Isoperimetric Factor, Elongation, Stochastic Convexity, Maximal Indentation Depth and Lobedness. Luciano D. S. Pacifico et. al. [24] used Centroid Contour Curve shape signature (Sha) as one of the features in their model. Most commonly used shape features were Aspect Ratio, Circularity and Solidity.

Leaf are available in various colors mainly green, orange, purple-brown and red. This difference in leaf color is majorly due to the presence of four different pigments. Majority of leaves are of green color. But still we can use color as a feature in differentiating plants. Different leaves



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have different range of green spectrum. This makes color as one of the important features for leaf classification. Majorly the color features used in papers were mean, standard deviation, skewness, and kurtosis of the color channel. These [22, 14, 3, 18, 4, 15, 6, 7] were the papers where color was used used as a feature.

Texture is another predominant feature which is helpful for a botanist to identify a plant's species. Leaves have various textures like shiny, leathery, hairy and smooth. Determining these textures is not very difficult for humans, but when it comes to computer vision, it becomes a challenge. Leaf texture is a spatial arrangement of color and its intensities in a region. Techniques like co-occurrence matrix, Fractals, Gabor filters, variations of wavelet transform, etc have been used for extracting texture from an image. The most commonly used technique for texture extraction is Gray Level Co-occurrence Matrix (GLCM). GLCM calculates the number of occurrences of a pair of pixels with specified value and in a specified spatial relationship. GLCM is used to extract features like Contrast, Correlation, Energy and Homogeneity. Ana C. Siravenha et. Al. [2] uses 2DDiscrete Wavelet Transform (DWT) and GLCM to extract 1865 texture features.

Leaves have unique structures called veins. A tree leaves have several types of veins. The longest one is called as midvein and the other veins are connected to mid vein and form a unique pattern. The pattern of veins is known as venation. The venation structure and pattern are unique qualities than can be useful for classification with the help of machine learning models. Manisha Amlekar et. al. [15] uses canny edge detection to find the venation of leaves. A.H. Kulkarni et. Al. [6] used morphological operations on gray scale image of leaf to extract three different types of venation features. G. Saleem et. Al. [8] used applied four disk-shaped morphological operators of radius 1–4 mm on leaf images to extract four different venation structures which are then used to calculate five different venation features.

Mojgansadat Mohtashamian et. Al. [25] proposed a new measure that can be calculated to show how much of a leaf is toothed in a given maple species called teeth indicator. Aimen Aakif et. Al. [9] proposed a new feature named Shape Defining Feature where vertical lines are drawn on the body of the leaf and end points of each lines are calculated. After which slope between the two endpoints is calculated to determine the shape of the leaf.

#### *2.3.2. Deep Learning Based*

Hand Crafted Feature extraction is a conventional method of extracting features in image classification. There are human limitations when extracting hand crafted features. There is only a limited number of features that can be extracted by using hand crafted feature extraction techniques which becomes a problem once there is a large number of different classes which won't be sufficient for classification and will produce poor results. To overcome this issue, Deep Learning Based feature extractors can be used. Deep Learning based features can be achieved by using a pre-trained Neural Network by allowing the input image to propagate forward and stopping at a pre-specified layer and using that layer as an output i.e. features. These features can then be further used for classification by feeding it to any classifier network.

S. Anubha Pearline et. Al. [7] utilizes VGG 16, VGG 19, Inception-v3 and Inception-ResNet-v2 pre-trained models as feature extractors. Four Publicly available datasets were used i.e. Folio, Swedish Leaf Dataset, Flavia and Leaf12 Dataset. VGG 19 performed best as a feature extractor paired with Logistic Regressor as a classifier.

### *2.4. Classification*

The previous sections discussed about various datasets, image preprocessing techniques and features of the image used in various research. In this section, with the help of the available data we compare all the methods and the classifiers that were used to classify different plant species. There has been various research where researchers have used publicly available dataset for plant classification. Flavia was the most commonly used dataset. Table 1 shows comparison of different



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Table 1: Comparison of Methods on Flavia Dataset

| Study | Feature Extraction Method | Classifier | Accuracy |
|---|---|---|---|
| [2] | 1865 Texture Features | ANN with 2 Hidden Layers | 91.85% |
| [3] | Shape, Texture, Color, Veins | SVM with Ant Colony Optimization | 96.25% |
| [4] | Deep CNN | Multi Layer Perceptron | 99.4 % |
| [5] | Hand Crafted Features | LeafNet CNN | 97.9% |
| [6] | Shape, Vein, Texture, Color and Zerkine Moment | Radial Basis Probabilistic Neural Network | 93.82% |
| [7] | VGG 19 | Logistic Regression | 96.25% |
| [8] | Shape, Venation, Statistical Features | K-Nearest Neighbor | 98.75% |
| [9] | Shape, Shape-Defining Feature, Fourier Descriptor | Multi Layer Neural Network with Back Propagation | 96% |
| [10] | Deep Learning Based | ResNet26 Model | 99.65% |

studies based on the classifier and feature extractor used on the Flavia Dataset.

Yu Sun et. al. [10] achieved the highest accuracy of 99.65% for Flavia dataset with their proposed ResNet26 Model. S. H. Lee et. al. [4] achieved the second-best accuracy of 99.4% for Flavia dataset with a Multi-Layer Perceptron.

Plant Seedling was the second most used Dataset used in [11, 12, 13]. Belal A . M. Ashqar et. Al. [11] used VGG-net 16 layers architecture on Plant Seedling dataset which achieved an accuracy of 99.48%. Catherine R. Alimboyong et. Al. [12] proposed a method which classifies Plant Seedling dataset with an AlexNet 25-layer network achieving an accuracy of 99.69%. The feature extraction method used in this study was Deep Learning based method. Carlos A. Mamani Diaz et. Al. [13] used three deep learning models namely Xception, VGG16, and Inception V3 which achieved an accuracy of 86.21%, 78.85% and 76.6% respectively.

### *2.5. Deployment*

After successful classification of plant species, the next step is to deploy the whole system into a user-friendly platform where the user can import an image of a plant and the system can classify the plant image. There are various platforms where a deep learning project can be deployed, like an Android Application, Website, Web-based service or a Desktop Application. These services can be offered in either offline or online as per the requirement. Deployment of a deep learning model is a challenging task since there are various variables which needs to be taken care. There are memory limitations, processing power limitations, cross platform compatibility issues, etc. David Knight et. Al. [23] built an android application which classifies 5 different leaves. MATLAB implementation performed better than the android application.

## 3. Findings

After analyzing and comparing 27 research papers we observed some points which are the findings of this study. They are as follows:

- **Real life datasets were not used:** Publicly datasets like Flavia, Folia, ICL and Swedish Leaf Dataset some of the widely used plant leaf datasets available on the internet. All of these datasets are of the Only Leaf type category i.e. only leaf is present in the image with no background. Training with such datasets makes it difficult for the classifier to work where the input data consists of noise, background and unwanted objects. There needs to be a method to train the model in such a way where it can work in real life examples. Plant seedlings is one of the datasets



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where noise, background and other unwanted objects are present in the image along with the leaf which makes the model robust.

- **Data Augmentation was not widely used:** Data Augmentation is one of the most popular and rapidly growing technique used to avoid overfitting and enlarging the dataset. Data Augmentation is a technique where data is generated by altering the orientation, size, color, horizontal and vertical shifting, image zooming, image flipping, rotation and other parameters to increase the size of the dataset. Only [11], [14], [20], [12], [21] researches in our review had used data augmentation techniques to increase the existing dataset. was the most commonly used part for classification: Most researches used leaf as their only organ to classify plants. There were a very few numbers of researches which used Multiple Organs to classify plants. Figure **??** shows the number of datasets which had multiple organs present in the image that can be used to classify plants. Ignacio Heredia [19] used collection of images available in PlantNet under a Creative-Common Attribution-ShareAlike 2.0 license. The image consisted habit, flower, leaf, bark and many more plant parts which were labelled by the experts.

- **Deep Learning based feature extraction performed better:** Hand Crafted Feature extraction techniques was majorly used for this problem. And as discussed in the Feature Extraction section, there are only a limited number of hand-crafted features that can be extracted. This creates a problem when you are dealing with a large dataset with a large number of species. To overcome this problem Deep Learning based feature extraction needs to be used. Anubha Pearline et. Al. [7] compared the performance and accuracy of classification with hand crafted features and deep learning-based feature extraction. This comparison was done on 4 publicly available dataset i.e. Flavia, Swedish Leaf, Leaf12 and Folio dataset. In all cases deep learning-based feature extraction yielded the best results.

- **Deployment was not implemented in studies:** Deployment is one of the most important process of any system. It helps a user to interact with the system effortlessly and efficiently. As discussed earlier, deployment is a challenging task and we found only single study [23] which had deployed their system in an Android application.

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